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Qualification Pack



Certificate Course in Coding Skills

QP Code: ASP/ITS/Q0201

Version: 1.0

NSQF Level: 5

Additional Skill Acquisition Programme Kerala || Additional Skill Acquisition Programme Kerala, 3rd
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Qualification Pack

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ASP/ITS/Q0201: Certificate Course in Coding Skills

Brief Job Description

Individuals at this job are responsible for developing applications and platforms in any language in order to build a robust hack free system. They will be responsible for evaluating the technical performance of algorithmic models on the system on which it is being deployed. They will be responsible for developing, designing, building, testing and deploying programming solutions.

Personal Attributes

Strong Programming Skills, Problem-Solving Skills, Analytical Skills, Understanding of Algorithms and Data Structures

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASP/ITS/N0201: Design Thinking](#)
2. [ASP/ITS/N0202: Logical Thinking](#)
3. [ASP/ITS/N0203: Programming.](#)
4. [ASP/ITS/N0204: Community Linking](#)
5. [ASP/ITS/N0205: Design of Software](#)
6. [ASP/ITS/N0206: Project Development](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	Programming
Occupation	Coding Skills
Country	India
NSQF Level	5

Qualification Pack

Credits	9
Aligned to NCO/ISCO/ISIC Code	2512.0800
Minimum Educational Qualification & Experience	Pursuing 3rd year of UG (Students pursuing Engineering Degree & should have completed 5 semesters.) with NA of experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	09/04/2025
NSQC Approval Date	09/04/2022
Version	1.0
Reference code on NQR	2020/ITES/ASAP/03802
NQR Version	1

Qualification Pack

ASP/ITS/N0201: Design Thinking

Description

The trainee will be aware of the latest trends in industry. Familiarization of tools and Techniques for design thinking, prototyping.

Scope

The scope covers the following :

- Comprehensive understanding of the design thinking and its practices.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. IR 4.0 Exploration and Mind mapping

PC2. Setting Up Platform Skills for Design Thinking & Orientation towards Sustainability

PC3. Design Thinking in Practice

PC4. Launchpad

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	20	-	10
PC1. IR 4.0 Exploration and Mind mapping	5	5	-	2
PC2. Setting Up Platform Skills for Design Thinking & Orientation towards Sustainability	5	5	-	2
PC3. Design Thinking in Practice	5	5	-	3
PC4. Launchpad	5	5	-	3
NOS Total	20	20	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0201
NOS Name	Design Thinking
Sector	Information Technology Sector
Sub-Sector	
Occupation	Coding Skills
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	09/04/2022
Next Review Date	09/04/2025
NSQC Clearance Date	09/04/2022

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ASP/ITS/N0202: Logical Thinking

Description

Familiarizing Problem Solving using puzzles. Using Roboquest the trainees will understand how to write basic if then else statement and loops plus testing.

Scope

The scope covers the following :

- Comprehensive understanding of logical puzzles & games and online interactive tools.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- | | |
|-------------|----------------------------------|
| PC1. | Logical Puzzles and Games |
| PC2. | Classroom activities (RoboQuest) |
| PC3. | Roboquest Part II |
| PC4. | Online Interactive Tool |

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	20	-	10
PC1. Logical Puzzles and Games	5	5	-	2
PC2. Classroom activities (RoboQuest)	5	5	-	2
PC3. Roboquest Part II	5	5	-	3
PC4. Online Interactive Tool	5	5	-	3
NOS Total	20	20	-	10

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0202
NOS Name	Logical Thinking
Sector	Information Technology Sector
Sub-Sector	
Occupation	Coding Skills
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	09/04/2022
Next Review Date	09/04/2025
NSQC Clearance Date	09/04/2022

Qualification Pack

ASP/ITS/N0203: Programming.

Description

Understanding evolution of programming. Creating interest in students by demonstrating projects

Scope

The scope covers the following :

- Comprehensive understanding of the Programming: The What, Why & How of Programming. Basics of SQL operations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Introduction to Programming: The What, Why & How of Programming

PC2. Exploring the possibilities

PC3. Dive into Programming - Python Basics & Application of Python.

PC4. Coding Artist

PC5. Introduction to database and SQL

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	25	25	-	10
PC1. Introduction to Programming: The What, Why & How of Programming	5	5	-	2
PC2. Exploring the possibilities	5	5	-	2
PC3. Dive into Programming - Python Basics & Application of Python.	5	5	-	2
PC4. Coding Artist	5	5	-	2
PC5. Introduction to database and SQL	5	5	-	2
NOS Total	25	25	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0203
NOS Name	Programming.
Sector	Information Technology Sector
Sub-Sector	
Occupation	Coding Skills
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	09/04/2022
Next Review Date	09/04/2025
NSQC Clearance Date	09/04/2022

Qualification Pack

ASP/ITS/N0204: Community Linking

Description

Trainee will understand the power of collaboration and its effectiveness. Trainee gain knowledge on various open source platform and solutions

Scope

The scope covers the following :

- Comprehensive understanding of the power of collaboration and explore various open sources.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- | | |
|-------------|-------------------------|
| PC1. | Power of Collaboration |
| PC2. | Open-Source Exploration |
| PC3. | GitHub Master |
| PC4. | The Helping Hands |

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	20	-	10
PC1. Power of Collaboration	5	5	-	2
PC2. Open-Source Exploration	5	5	-	2
PC3. GitHub Master	5	5	-	3
PC4. The Helping Hands	5	5	-	3
NOS Total	20	20	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0204
NOS Name	Community Linking
Sector	Information Technology Sector
Sub-Sector	
Occupation	Coding Skills
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	09/04/2022
Next Review Date	09/04/2025
NSQC Clearance Date	09/04/2022

Qualification Pack

ASP/ITS/N0205: Design of Software

Description

Understand the need of software engineering and how big companies manage the development using Agile.

Scope

The scope covers the following :

- Understanding the importance of Software Designing and how big companies does the magic.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Design of Software Understanding the importance of Software Designing.

PC2. Waterfall vs Agile

PC3. Software Testing and Project Management tools.

PC4. Agile Development Demo

PC5. Project Idea

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	25	25	-	10
PC1. Design of Software Understanding the importance of Software Designing.	5	5	-	2
PC2. Waterfall vs Agile	5	5	-	2
PC3. Software Testing and Project Management tools.	5	5	-	2
PC4. Agile Development Demo	5	5	-	2
PC5. Project Idea	5	5	-	2
NOS Total	25	25	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0205
NOS Name	Design of Software
Sector	Information Technology Sector
Sub-Sector	
Occupation	Coding Skills
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	09/04/2022
Next Review Date	09/04/2025
NSQC Clearance Date	09/04/2022

Qualification Pack

ASP/ITS/N0206: Project Development

Description

Project Planning and Work Division - Team should convert the project idea to proper project plan using tools. Project Requirement and Design - A requirement document/backlog creation and design documents. Implementation & Testing - Build creating version history, automated testing /devops. Project Presentation - Project demo and presentation.

Scope

The scope covers the following :

- Conversion of the project idea to proper project plan using tools.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- | | |
|-------------|------------------------------------|
| PC1. | Project Planning and Work Division |
| PC2. | Project Requirement and Design |
| PC3. | Implementation & Testing |
| PC4. | Project Presentation |

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	60	60	-	10
PC1. Project Planning and Work Division	10	10	-	2
PC2. Project Requirement and Design	10	10	-	2
PC3. Implementation & Testing	10	10	-	3
PC4. Project Presentation	30	30	-	3
NOS Total	60	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0206
NOS Name	Project Development
Sector	Information Technology Sector
Sub-Sector	
Occupation	Coding Skills
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	09/04/2022
Next Review Date	09/04/2025
NSQC Clearance Date	09/04/2022

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment Strategy

ASAP follows an assessment framework which provides weightage for all the activities connected with skilling in which students get involved during the training program. The components of assessment include Attendance, Assignments, Internal Assessments and Final Assessment. Of these, Attendance and Internal Assessment come under Continuous and Comprehensive Evaluation (CCE). All Assessments with regard to the academic status of the student should be reflected as marks and overall assessment by awarding grades. The report card/certificate will state only the overall grade.

WEIGHTAGES

For IT-ITeS and similar sector courses (knowledge based) aligned to NSQF at Level 5 and above, the assessment shall be done with 60% weightage for practical component and 40% weightage for theoretical

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part. For other courses where NSQF Levels are 4 and below, 70% weightage shall be given for practical component and 30% weightage shall be given for theoretical knowledge.

ATTENDANCE CRITERIA

Skill based sessions and training are delivered through lectures, discussions, demonstrations, and experiments ensuring student participation during daily learning activities. So those who miss such sessions will not be able to compensate for the loss. Regular student participation in daily classroom activities plays a significant role in student's success. For each course and batch, the student's attendance will be marked daily, and marks will be awarded accordingly on course completion before final assessment. Students whose attendance falls below 70% will not be eligible for final assessment and course completion certificate.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASP/ITS/N0201.Design Thinking	20	20	0	10	50	10
ASP/ITS/N0202.Logical Thinking	20	20	0	10	50	10
ASP/ITS/N0203.Programming.	25	25	0	10	60	20
ASP/ITS/N0204.Community Linking	20	20	0	10	50	10
ASP/ITS/N0205.Design of Software	25	25	0	10	60	20
ASP/ITS/N0206.Project Development	60	60	0	10	130	25
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	190	200	-	60	450	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.